

Colophon

****Title:** Towers YM v2.3 – H4 Boundary Protocol**
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****Version:** 2.3 FINAL**

****P5_genuine = 1000000001119 [13 digits, Sym=1] LOCKED****

Build protocol

****Formal + Empirical****
****Axioms:** 0**
****Sorry:** 0**
****Mathlib:** OFF (mathlib-free, Lean core only)**
****Provenance:** ON**

Modules

On-disk filenames use underscores; the published module names are listed first.

1. ****H4Core**** (`Towers/YM/H4Core.lean` + leaf `Towers/YM/H4\_Strata\_Ztau.lean`) – Task #324 – manifest SHA *record* `48536d9d...ba997` (prose milestone record hash, NOT a file hash). Real file SHA-256 `aa8c1180...a6ce`, recorded in `H4Core.data.json`.
2. ****H4Boundary**** (`Towers/YM/H4\_Boundary.lean`) – Task #325 – `C13 := 13`. ****Certified 13-character P5 table**** (digit counts tool-certified via `{#literal}`, never eye-counted):

Name	Value	Digits	Sym	Status
P5_genuine	1000000001119	13	1	**CERTIFIED · LOCKED**
P5_proposed_14	10000000001119	14	–	REJECTED
P5_proposed_15	100000000001119	15	–	REJECTED

`P5\_genuine = 1000000001119` is the ONLY live P5 in v2.3 – the real 13-digit boundary prime (`digit\_len = 13`, `Sym = 1`). The two over-zero variants each carry one extra zero, so their `digit\_len P5 = C13` (= 13) claim is ****FALSE****; they are kept solely as labeled ****REJECTED**** counter-records, not used as P5 anywhere. `H4\_Boundary.data.json` records the 14-digit `10000000001119` as the honest counter-record. No spike, transition, or `Sym`/critical-line meaning is attached to digit length (see `docs/BOUNDARY\_ARTIFACT.md`).
3. ****H4TimeBound**** (`Towers/YM/H4\_TimeBound.lean`) – Task #326 – `N40 := 40`; `TimeHorizon := 3^40` carries no causal/temporal meaning; `10^12 < 3^40` (by ~7 orders of magnitude).
4. ****H4Derivation**** (`Towers/YM/H4\_Derivation.lean`) – Task #327 – file SHA-256 `e21aelba...e2b2`. `C13\_Law\_Open`: ****CONJECTURE**** (named open `Prop`; no theorem discharges it).

Core results

****Primary law (CONJECTURE – empirical 6/6 only, NOT proven):****
`digit\_len p ≥ 13 → Sym p = 1`.

****Demoted:**** `N40 := 40`. `TimeHorizon := 3^40` is not causal – a plain integer the witnesses are compared against.

****Computed (real W(H4) action, not hardcoded):**** `Sym(19) = 2`, `Sym(191) = 2`,
`Sym(1000000001119) = 1`. (`Sym(19) = 2`, NOT 20.)

Z Experiment Harness (v2.3)

****Tool use eliminates digit reproduction error.**** Pure measurement, no conjecture: 50 script-generated inputs (digit length 10–25, zero-run 0–15,

`Sym ∈ {1, 2, null}`) were reproduced tool-assisted (`printf` / `\${#s}`) over 5000 trials (100 per input). Result: **0 errors, error rate 0.0** – tool-assisted reproduction is deterministic across every tested input.

Scope (honest): this measures only the tool-assisted regime (`T = 1`). With all errors zero there is no correlation between reproduction error and `Sym`, digit length, or zero-run when `T = 1` – but a null result cannot rank those factors, and the harness does NOT test the pure-generation (`T = 0`) regime. No `Sym`-coupling, digit-band, or "metastability" law is implied. Harness: `experiments/z-metastability/`.

Gates

Gematria: OFF | Abjad: OFF | Religious: OFF | AI Attribution: OFF

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